ROBUST AND RELIABLE EARLY-STAGE WEBSITE FINGERPRINTING ATTACKS VIA SPATIAL-TEMPORAL DISTRIBUTION ANALYSIS

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CNS:Project

Introduction

- Website Fingerprinting is a cyber attack technique used to identify which website a user is visiting by analyzing encrypted internet traffic.
- Even though Tor encrypts the data, attackers can still study:
- packet direction
- timing
- traffic patterns
- This project uses Deep Learning to identify websites from encrypted traffic.

Problem Statement

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Existing Website Fingerprinting attacks:

- need complete traffic data
- fail under network changes
- perform poorly with defenses
- cannot detect websites early

The goal of this project is to detect websites during the early stage of page loading.

Objective

- Understand Website Fingerprinting attacks
- Analyze encrypted traffic patterns
- Train Deep Learning models
- Detect websites using early traffic
- Test attack performance on datasets

What is Holmes?

Holmes is a Deep Learning based Website Fingerprinting attack.
It uses:

- Spatial Analysis
- Temporal Analysis
- Contrastive Learning

Holmes can identify which website a user is visiting by analyzing network traffic patterns, even before the webpage fully loads.

Technology Stack

Technology	Purpose
Python 3.8	Used for programming
PyTorch	Used to build deep learning models
NumPy	Used for data processing
Scikit-learn	Used for dataset splitting
VS Code	Used for writing and running code
WSL Ubuntu	Used as a Linux environment

Dataset Used

Dataset Information

Dataset Used:

- NCDrift_inf.npz

Dataset Details:

- 93 website classes
- 6882 traffic samples
- sequence length = 5000

Traffic data contains:

- packet direction
- packet timing information

Dataset Splitting

The dataset was divided into:

Dataset	Samples
Training	5573
Validation	620
Testing	689

Purpose:

1. **Training → model learning**
2. **Validation → tuning**
3. **Testing → final evaluation**

Project Workflow

Traffic Dataset

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Preprocessing

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Dataset Splitting

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Feature Extraction

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DF Model Training

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Testing & Evaluation

Model Used

Deep Fingerprinting (DF)

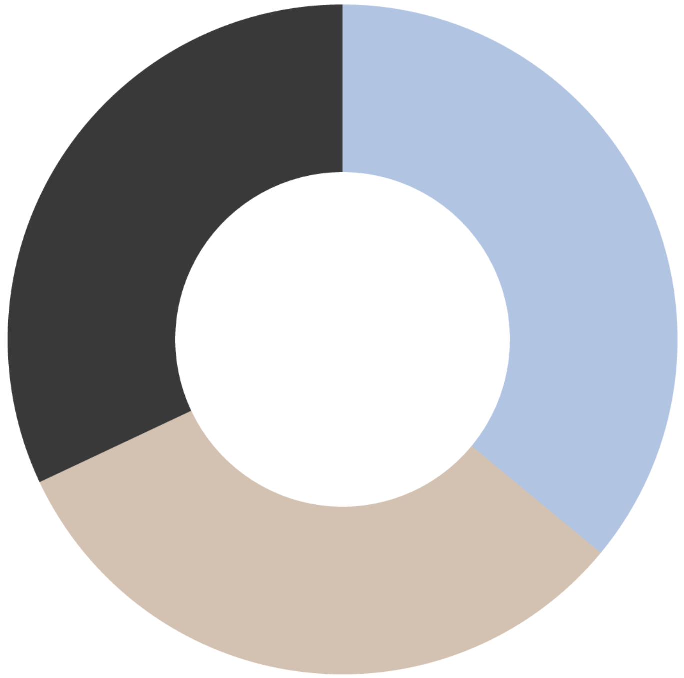
The project uses the DF model.

DF is:

- a CNN-based Deep Learning model
- designed for Website Fingerprinting attacks

Functions:

- extracts traffic patterns
- learns encrypted traffic behavior
- classifies websites



Training Process

Steps performed:

- loaded dataset
- split dataset
- trained DF model
- generated checkpoint file

Generated Model File:

max_f1.pth

Evaluation Metrics

The following metrics were used:

- Accuracy
- Precision
- Recall
- F1-score

These metrics help measure model performance.

Final Results

Accuracy: 47.61%

Precision: 21.75%

Recall: 25.86%

F1-score: 20.89%

The model successfully classified encrypted traffic patterns.

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Thank You